FitBuddy: AI Fitness Plan Generator using Gemini Models
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Project Planning Phase – FitBuddy

The **Project Planning Phase** focuses on organizing the development activities, technologies, milestones, and deployment steps required to develop the FitBuddy application.

1. Project Objectives

- Develop an AI-based personalized fitness plan generator.
- Generate a structured **7-day workout plan**.
- Provide nutrition or recovery tips.
- Allow users to update their plans through feedback.
- Store user and workout-plan information.
- Provide an admin view for monitoring users and plans.

2. Technology Planning

The project uses:

- **Python** – Programming language
- **FastAPI** – Backend framework
- **Google Gemini** – AI models
- **HTML, CSS, Jinja2** – Frontend
- **SQLite + SQLAlchemy** – Database
- **Uvicorn** – Application server
- **Git** – Version control

3. Development Milestones

Milestone 1 – Model Selection and Architecture

- Research and select the appropriate Gemini model.
- Define the application architecture.
- Set up the development environment.

Milestone 2 – Core Functionality Development

- Develop workout generation, nutrition tips, feedback updating, and database functionality.
- Implement the FastAPI backend for routing and user-input processing.

Milestone 3 – routes.py Development

- Implement the main application logic and route handlers.

Milestone 4 – Frontend Development

- Design the user interface.
- Create dynamic Jinja2 templates.

Milestone 5 – Deployment

- Prepare the application for local deployment.
- Test and verify the local deployment.

4. Project Workflow

Requirement Analysis → Model Selection → Architecture Design → Backend Development → AI Integration → Frontend Development → Database Integration → Testing → Local Deployment

5. Testing and Deployment Plan

The application is planned to run locally using **Uvicorn**. The development server can be accessed through `127.0.0.1:8000`, with FastAPI's `/docs` interface available for API testing.

6. Expected Outcome

The planned outcome is a complete FitBuddy web application that provides **personalized fitness plans, nutrition guidance, feedback-based plan updates, and an administrative user view**.

Continue planning FitBuddy

- Create a FitBuddy project timeline
- Define FitBuddy testing activities